

08.26.26

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Times Series **Bitcoin Modeling**



Agenda

01 Introduction

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03 Modeling

04 Recommendations and Conclusions

05 Next Steps

01

Introduction



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Bitcoin Analysis



Hedge fund is seeking alternative asset classes for its investors. The hedge has observed that retail and institutional investors have generated positive returns via cryptocurrency. As a result, the hedge fund is tasking KBO Analytics to create a model that can predict Bitcoin prices.

Modeling.....

Completed the following:

- Created several models to predict Bitcoin prices
- Identified the best performing model

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02

Data



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Data Description

- csv file
 - Initially 4,532 rows of data
 - Total of 8 Columns
- Columns contain, but not limited, to the following:
 - *Start* (Date)
 - *End* (Date)
 - *Open* (Price)
 - *High* (Price)
 - *Low* (Price)
 - *Close* (Price)

03

Modeling



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Time Series

- What is a Time Series?
 - “.....model that can analyze sequential time series data and predict future values.....”



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Model Comparisons

- Utilized Root Mean Square Error (RMSE) to measure model performance
 - Measure of the distance between the predicted values and actual values
 - the smaller the RMSE error, the better the model performance

#	Model	RMSE
1	Naive Forecast	0.0222
2	Moving Average	2.6584
3	ARIMA	25719.2278
4	Auto-ARIMA	0.4469

04

Recommendations and Conclusions



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Recommendations and Conclusions

- Naïve Forecast is the best Time Series Model
 - Bitcoin Data behaves like a Random Walk
 - “.....*past price action has little or no influence on future changes.....*”

05

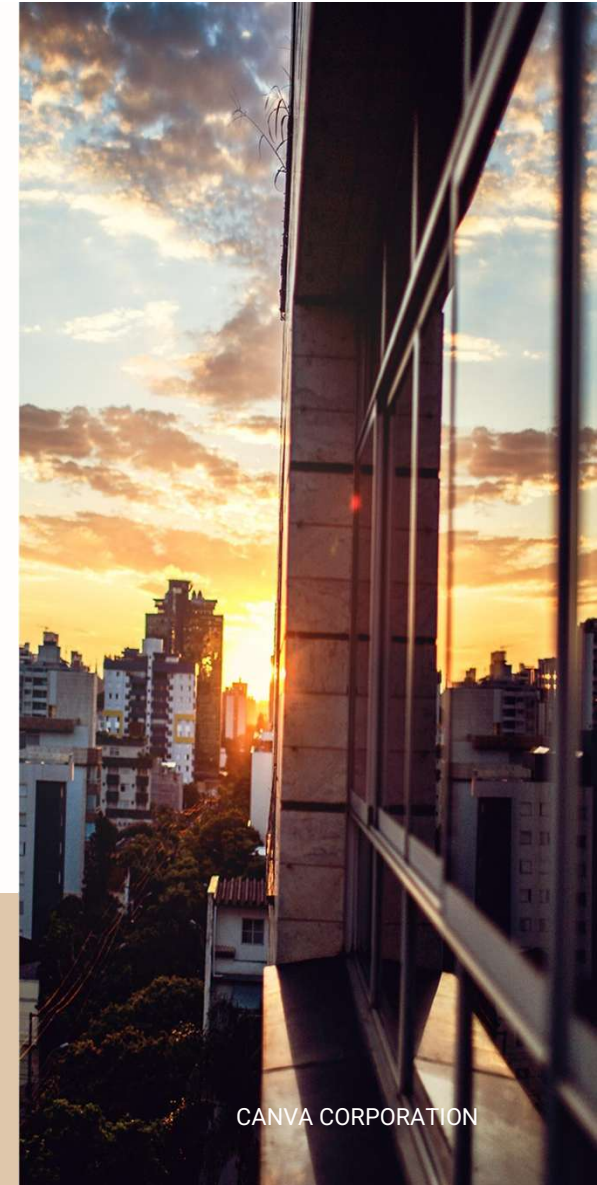
Next Steps



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Next Steps

- 1 – Additional Data
- 2 – Institutional Investment
- 3 – Other Cryptocurrencies



06

Appendix



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Appendix

- Belcic, Ivan and Cole Stryker. (2026). “What is a time series model?”
- Smith, Tim. (2026). “Exploring Random Walk Theory: Definition, Usage, and Insights.”